Sheet 37 Thu 28 Jan. 2027 *Quotient equation* $\frac{A(x)}{B(x)} = k$

1. Warm-up · Quick [EQ] For which value of x is the expression $\dfrac{1}{x-3}$ undefined?

   Ans. $x = 3$.

2. Core · Method [EQ] Solve $\dfrac{2x-1}{x+3} = 0$.

   Ans. $x \neq -3$ and $2x - 1 = 0$: $x = \dfrac{1}{2}$.

3. Core + · Method [EQ] Solve $\dfrac{x+1}{x-2} = 3$.

   Ans. $x \neq 2$; $x + 1 = 3(x-2)$, that is $x = \dfrac{7}{2}$. Check: $\dfrac{4.5}{1.5} = 3$.

4. Review · Expand [DEV] Expand and simplify $(x+2)(x-5)$.

   Ans. $x^2 - 3x - 10$.

5. Thread · Code reading [ALG] What do $g(3)$ and $g(-5)$ return?

```
def g(x):
    if x > 0:
        return 2*x
    else:
        return -x
```

   Ans. $g(3) = 6$ and $g(-5) = 5$.

   The condition is evaluated before the calculation: the branch depends on the sign of x.

Sheet 38 Mon 1st Feb. 2027 *Sign of a product or quotient; sign tables*

1. Warm-up · Quick [EQ] Sign of $2x - 4$?

   Ans. Negative for $x < 2$, zero at 2, positive for $x > 2$.

2. Core · Method [EQ] Draw the sign table of $(x-1)(x+3)$.

   Ans. Roots -3 and 1: positive on $]-\infty;-3[$, negative on $]-3;1[$, positive on $]1;+\infty[$.

   Sign of the product: rule of signs, column by column.

3. Core + · Method [EQ] Solve $\dfrac{x+2}{x-1} \geqslant 0$.

   Ans. Numerator positive for $x \geqslant -2$, denominator positive for $x > 1$; $S =]-\infty;-2] \cup]1;+\infty[$ (1 is excluded).

4. Review · Quick [PRB] Pack of 32 cards: probability of drawing a heart? a king?

   Ans. $\dfrac{8}{32} = \dfrac{1}{4}$; $\dfrac{4}{32} = \dfrac{1}{8}$.

5. Thread · Disprove [LIT] Disprove: "for every real number x, $x^2 \geqslant x$".

   Ans. Counterexample: $x = \dfrac{1}{2}$: $x^2 = \dfrac{1}{4} < \dfrac{1}{2} = x$.

   A single counterexample is enough to disprove a universal statement.

Proportions and percentage change

Sheet 39 Tue 2 Feb. 2027 *Proportions and percentages; proportion of a proportion*

1. Warm-up · Quick [PRO] Calculate 15 % of 240.

   Ans. $0.15 \times 240 = 36$.

2. Core · Method [PRO] 30 % of the pupils are boarders and 40 % of the boarders do dance. What proportion of the pupils are boarders who do dance?

   Ans. $0.3 \times 0.4 = 0.12 = 12\,\%$: proportion of a proportion (we multiply).

3. Core + · Problem [PRO] A school has 800 pupils, of whom 55 % are girls; 20 % of the girls study Latin. How many girls study Latin?

Ans. $800 \times 0.55 = 440$ girls; $440 \times 0.20 = 88$.

4. Review · Quick [QP] 3 exercise books cost 4.50 €. Price of 7 books?
   Ans. $1.50 \times 7 = 10.50$ €.
5. Thread · True/false [EQ] True or false? Justify. The number -2 is a solution of the equation $3x + 1 = -5$.
   Ans. True. $3 \times (-2) + 1 = -6 + 1 = -5$.

Sheet 40 Thu 4 Feb. 2027 *Absolute change, multiplier, rate of change*

1. Warm-up · Quick [PRO] A price goes from 80 Ar to 92 Ar. What is the absolute change?
   Ans. $92 - 80 = 12$ Ar.
2. Core · Method [PRO] For the same change, give the rate of change and the multiplier.
   Ans. $t = \dfrac{12}{80} = 0.15 = 15\,\%$; $CM = 1 + t = 1.15$.
3. Core + · MCQ [PRO] A decrease of 20 % corresponds to the multiplier:
   A. 0.2 **B.** -0.8 **C.** 1.2 **D.** 0.8
   Ans. D. 0.8
   $CM = 1 + t = 1 - 0.20$.
4. Review · Quick [STA] Temperatures -2, 0, 3, 5, 9 (in °C). Mean and range?
   Ans. Mean 3 °C; range $9 - (-2) = 11$ °C.
5. Thread · Quick [SOL] Calculate the volume and the total surface area of a cube with edge 3 cm.
   Ans. $V = 3^3 = 27$ cm^3; $A = 6 \times 3^2 = 54$ cm^2.

Sheet 41 Mon 8 Feb. 2027 *Successive percentage changes*

1. Warm-up · Quick [CAL] Calculate 1.2×0.9.
   Ans. 1.08.
2. Core · Method [PRO] An increase of 20 % is followed by a decrease of 10 %. What is the overall rate of change?
   Ans. $1.2 \times 0.9 = 1.08$: overall increase of 8 %.
   Rates are not added; it is the multipliers that are multiplied.
3. Core + · True/false [PRO] True or false? Justify. An increase of 50 % followed by a decrease of 50 % returns to the initial value.
   Ans. False. $1.5 \times 0.5 = 0.75$: overall decrease of 25 %.
4. Review · Quick [SUB] Calculate $x^2 - x$ for $x = \dfrac{1}{2}$.
   Ans. $-\dfrac{1}{4}$.
5. Thread · Write-up [DEM] $ABCD$ is a parallelogram such that $AC = BD$. What type of quadrilateral is it? Justify in one sentence.
   Ans. Rectangle: a parallelogram whose diagonals have equal length is a rectangle.

Sheet 42 Tue 9 Feb. 2027 *Reverse percentage change; proportion and change*

1. Warm-up · Quick [CAL] Calculate $\dfrac{1}{1.25}$.
   Ans. 0.8.
2. Core · Method [PRO] After an increase of 25 %, what decrease brings back the initial value?
   Ans. $CM_{\text{rev}} = \dfrac{1}{1.25} = 0.8$: decrease of 20 % (not 25 %).
   Pitfall: believing that an increase of 25 % is cancelled by a decrease of 25 %.
3. Core + · Interpret [PRO] Out of 250 candidates, 150 are successful; last year there were 200 candidates. Calculate the proportion of successful candidates, then the rate of change of the number of